Math 142, S14 | Name: Answer
Quiz 2 Section: | SSN: xxx-xx-

Instructions: There ars totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Evaluate the following integrals using *integration by parts*:

(a) $\int \ln(2x)\, dx$

Let $u = \ln(2x)$, $dv = dx$
$du = \frac{1}{x}dx$, $v = x$
then

$$\int \ln(2x)dx = x\ln(2x) - \int x \cdot \frac{1}{x} dx$$
$$= x\ln(2x) - \int dx$$
$$= x\ln(2x) - x + C$$

(b) $\int_0^1 te^{-t}\, dt$

Let $u = t$, $dv = e^{-t}dt$.
$du = dt$, $v = -e^{-t}$
then

$$\int_0^1 te^{-t}dt = (-te^{-t})\Big|_{t=0}^{t=1} - \int_0^1 (-e^{-t})dt$$
$$= (-e^{-1}) - (e^{-t})\Big|_{t=0}^{t=1}$$
$$= -e^{-1} - (e^{-1} - 1)$$
$$= 1 - 2e^{-1}$$

Problem 2. (4 points) Use appropriate substitution to find $\int \cos^2 x \sin^3 x\, dx$.

Let $u = \cos x$, then $du = -\sin x\, dx$

$$\int \cos^2 x \sin^3 x\, dx = \int \cos^2 x \sin^2 x \sin x\, dx$$
$$= \int \cos^2 x (1 - \cos^2 x) \sin x\, dx$$
$$\overset{u=\cos x}{=\!=} \int u^2(1-u^2)(-du)$$
$$= \int u^4 - u^2\, du$$
$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$
$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$